

JAYCE HILL

Principal AI Security Architect

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EXECUTIVE SUMMARY

Principal AI Security Architect with 20+ years leading enterprise security architecture, governance, and risk management across Fortune 100 organizations in financial services, energy, automotive, and capital markets. Specializes in translating complex regulatory requirements and organizational risk into practical, scalable security frameworks that protect AI and cloud deployments without slowing delivery. At Microsoft ISE, served as forward deployed security architect across ~20 strategic engagements — partnering directly with CISOs, compliance leads, and engineering organizations to align security policy, governance controls, and technical execution. Proven ability to operate across the executive and engineering layers: equally comfortable presenting to senior leadership and shipping hardened infrastructure. Previously held TS/SCI CI-Poly Full Scope Lifestyle clearance.

CORE COMPETENCIES

- Enterprise Security Architecture & Governance
- AI Security Strategy for Regulated Industries
- Risk Management & Regulatory Compliance (RMF, NIST, FedRAMP)
- Zero Trust Architecture & Identity Governance
- Cloud Security (Azure, AWS, OCI) for Regulated Environments
- Cross-functional Stakeholder Alignment (CISO to Engineering)
- Software Supply Chain Security & DevSecOps
- Secure-by-Default Infrastructure (IaC, Policy-as-Code)
- Threat Modeling (CRISP, STRIDE, MITRE ATT&CK)
- Security Program Development & Customer Enablement
- Governance Automation & Audit Traceability
- Executive Communication & Security Culture Leadership

TECHNICAL SKILLS

Platforms: Azure (Solutions Architect Expert), AWS, Oracle Cloud Infrastructure

Languages: Python, C#, PowerShell, Bash, Bicep

Governance & Compliance: NIST 800-53, RMF, FedRAMP, OpenSSF Scorecard, SLSA, Zero Trust Architecture

Security Tooling: CRISP MCP, Defender for Cloud, Sentinel, CodeQL, Trivy, CycloneDX, GitHub Advanced Security

Frameworks: OWASP, MITRE ATT&CK, STRIDE, SBOM/Supply Chain, IaC Security

PROFESSIONAL EXPERIENCE

Principal AI Security Architect | Microsoft — Industry Solutions Engineering 2021 – Present

Forward deployed security architect partnering with Fortune 100 and public sector customers across financial services, energy, automotive, and capital markets. Operated as principal-level technical lead and trusted advisor to customer CISOs and engineering organizations. Level 65 (Principal).

- Led security architecture and governance strategy across ~20 Fortune 100 engagements including Bank of America, Fidelity, LSEG, Halliburton, Dow, General Motors, KPMG, and FIS — translating regulatory requirements and business risk into actionable engineering controls and security roadmaps.
- Co-led the build and deployment of CRISP, a proprietary threat modeling and security architecture framework, directly within customer engineering sprints. Identified by ISE leadership as a top-priority accelerator; adopted as a standard delivery asset across the ISE organization.
- Served as a key Security Lead for HVE-Core; an Agentic engineering platform with 34 autonomous agents executing Research-Plan-Implement workflows via GitHub Copilot, available as a public repository on Microsoft's GitHub. Led customer deployment workshops at KPMG and LSEG.
- Designed and delivered a repeatable AI security governance framework for Halliburton/Helios — a multi-tenant AI platform processing proprietary oil & gas data across regulated Azure environments — enabling continuous security validation as architecture evolved weekly without becoming a delivery bottleneck.
- Produced customer-specific security gap analyses for KPMG, FIS, and Halliburton with full regulatory traceability: mapping regulation to risk gap to engineering control to implementation artifact. Framework adopted as a reusable methodology across ISE engagements.

- Co-led security breakout sessions at KPMG workshop (New Jersey), facilitating senior leadership discussions on integrating security governance into AI delivery workflows and building an inclusive, scalable security culture.
- Built a secure-by-default infrastructure baseline spanning 11 Azure services (AKS, SQL, Cosmos, Key Vault, Redis, AI Foundry, Storage, VNet, NSG, PIM, Guardrails) with Zero Trust defaults enforced at the IaC layer — reducing misconfiguration risk and accelerating compliant customer onboarding.
- Hardened CI/CD and software supply chain controls across ISE accelerator platforms, implementing SBOM generation, artifact signing, container scanning, and workflow permission governance aligned to OpenSSF Scorecard standards.
- Traveled to LSEG London for the HVE Hackathon; led Solution Architecture Design process investigation, produced an 8-section SAD Submission Readiness Checklist adopted across engagements, and presented findings at the Science Fair.
- Operated cross-functionally with customer CISOs, compliance teams, and engineering leads to align security policy, operational constraints, and technical execution — ensuring security acted as a delivery accelerator rather than a governance blocker.

Information System Security Manager | General Dynamics IT (GDIT)

2020 – 2021

Security operations and compliance oversight in classified and regulated environments. TS/SCI CI-Poly Full Scope Lifestyle clearance supporting IC missions.

- Managed security accreditation (ATO) and RMF activities for classified and cloud-based systems across AWS and Azure, coordinating technical, compliance, and stakeholder workstreams simultaneously.
- Conducted architecture reviews and Technical Exchange Meetings with program leadership to assess risk posture, define security controls, and align remediation milestones across IC programs.
- Developed and executed cyber security improvement strategies including post-incident analysis for data spills and zero-day vulnerability resolution in regulated environments.

Senior Principal Consultant — National Security Group | Oracle

2011 – 2020

Lead security consultant across a 9-year engagement — OCI architecture, penetration testing, compliance governance, and executive program leadership for national security customers.

- Served as Oracle Cloud Infrastructure Solution Architect, designing cloud deployment architectures and leading certification and accreditation frameworks for national security programs.
- Facilitated quarterly stakeholder reviews and bi-monthly Technical Exchange Meetings with mission partners covering performance, risk management, emerging technology, and program milestones.
- Led end-to-end penetration testing, vulnerability assessments, and system hardening using MITRE ATT&CK across Oracle Exadata, Big Data Appliance, WebLogic, and Analytics platforms.
- Produced authoritative technical deliverables — System Security Plans (SSPs), CONOPs, whitepapers, and risk matrices — guiding program-level accreditation decisions and executive stakeholder alignment.
- Managed full project lifecycle including requirements gathering, architecture review, security testing, and end-to-end solution delivery within complex, customized cybersecurity environments.

EDUCATION

M.S. Cybersecurity (in progress) — Georgia Institute of Technology

M.Eng., Systems Engineering — George Washington University

B.S., Electrical & Computer Engineering — Michigan State University

CERTIFICATIONS & SECURITY CLEARANCE

- Microsoft Azure Solutions Architect Expert
- GIAC Security Essentials (GSEC)
- Certified Ethical Hacker (CEH)
- HashiCorp Terraform Associate
- TS/SCI CI-Poly — Full Scope Lifestyle (Previously Held; Reinstatement Eligible)

SELECTED CLIENT ENGAGEMENTS

Fortune 100 & Strategic Accounts: Bank of America · Fidelity · LSEG · KPMG · FIS · Halliburton (Helios) · Dow · General Motors · U.S. Federal & IC Agencies